

Top-Level Workplan – BFrep Manuscript Revision

July 2026

1 Top-Level Workplan – BFrep Manuscript Revision

1.1 Overview

This document coordinates three parallel work streams for salvaging the manuscript “A pluralist perspective on input-output modeling: searching for commensurability” (Kapeller & Scharnreitner, submitted to *Metroeconomica*).

The revision is based on the finding that the existing code base ((3)BeyondHulten) is approximately 70–80% functional – far more advanced than earlier assessments assumed. This reduces the total timeline from 12–16 weeks to 6–10 weeks.

1.2 Work Streams

1.2.1 Stream A: Manuscript Revision

Location: (1)Submission/revise/

Status: Scaffold created. The manuscript has been reorganized into a modular structure:

```
revise/  
---- main.tex           -- new main file with revised abstract  
---- BandF.bib          -- bibliography (copied)  
---- pictures/          -- figures (copied)  
---- chapters/  
    ---- ch01_introduction.tex -- % OLD TEXT FROM HERE ON  
    ---- ch02_io_leontief.tex  -- % OLD TEXT FROM HERE ON  
    ---- ch03_cge_model.tex    -- % OLD TEXT FROM HERE ON  
    ---- ch04_data.tex         -- % OLD TEXT FROM HERE ON  
    ---- ch05_ces_functions.tex -- % OLD TEXT FROM HERE ON  
    ---- ch06_labor_slack.tex  -- % OLD TEXT FROM HERE ON  
    ---- ch07_conclusion.tex    -- % OLD TEXT FROM HERE ON  
    ---- ch08_backmatter.tex   -- % OLD TEXT FROM HERE ON
```

The new abstract replaces the “paradigmatic cleavage” framing with a “labor supply elasticity continuum” framing, centered on Robinson (2006) and the quantitative decomposition of closure choice vs. technology choice.

Remaining work:

- Rewrite introduction (drop Kuhnian preamble to one paragraph, open with policy question, state upfront that IO is a special case of CGE)
- Rewrite Section 2 (streamline IO exposition, fix sign error in eq. 19, remove “supply-side vs demand-side” distinction, be transparent about open economy)
- Rewrite Section 4 (drop CES novelty claim, drop Cobb-Douglas as “archetypical neoclassical”)
- Replace Section 5 (exogenous labor slack -> endogenous labor supply with η parameter)
- Rewrite Section 6 (drop “deus ex machina” hand-wringing, present variance decomposition results)
- Integrate new literature: Robinson (2006), Rose (1995), Dervis et al. (1982), de Melo & Tarr (1992), McGregor et al. (1996), Shoven & Whalley (1984)
- Fix minor issues (p.5 sentence, p.16 paragraph, Figure 3 axis)

Dependencies: Requires results from Stream B (η sweep, variance decomposition) before Sections 5–6 can be finalized.

1.2.2 Stream B: Model Extension (BeyondHulten code base)

Location: (3)BeyondHulten/src/

Status: Core code is 70–80% functional. Three model types (CES, Leontief, Cobb-Douglas) work. German data is connected. Labor slack is implemented as a function parameter. Elasticity gradient and plotting infrastructure exist.

Remaining work:

- Implement continuous labor supply elasticity η in $[0, \infty)$ as a new labor slack function
- Switch from sector-specific to mobile labor with economy-wide wage
- Open the economy (imports, exports, domestic vs. imported intermediates) or document the closed-economy mapping transparently
- Implement variance decomposition ($\eta \times \varepsilon \times \theta \times \sigma$ factorial with partial R^2 or Sobol indices)
- Fix calibration documentation (summary table of α , β , λ , Domar weights, elasticities)
- Update notebooks to match current API (some notebooks use older function signatures)
- Go/no-go decision: run pilot η sweep, check whether sectoral results show interesting variation

Dependencies: Can proceed immediately on existing code. The B&F replication (Stream C) provides validation but is not a prerequisite.

1.2.3 Stream C: B&F Replication

Location: (3)BeyondHulten/bf_replication/

Status: Workplan created (REPLICATION_WORKPLAN.md). No code yet.

Purpose: A clean-room replication of Baqaee & Farhi (2019) on US data. Serves two functions:

- Validates the model implementation against published results
- Provides the computational infrastructure for the “replicate and contextualize” strategy from the salvage plan

Remaining work: See bf_replication/REPLICATION_WORKPLAN.md for the 6-phase plan (R1–R6).

Dependencies: Can reuse code patterns from BeyondHulten/src/. Independent of Streams A and B.

1.3 Cross-References

This workplan	References
Full salvage strategy	(4)plans/REVISED_SALVAGE_PLAN.md
Earlier plans (superseded)	(4)plans/SALVAGE_PLAN.md, salvage_plan2.md, salvage1_comment.md, BF_CONTRIBUTION_RECEPTION.md
Replication details	(3)BeyondHulten/bf_replication/REPLICATION_WORKPLAN.md
Manuscript source	(1)Submission/revise/main.tex + chapters/
Original manuscript	(1)Submission/foundations_of_pdf/submission(1).tex
Reviews	(2)Reviews/metro-rev1.docx, metro-rev2.docx
Core code	(3)BeyondHulten/src/

1.4 Execution Order

1.4.1 Weeks 1–2: Foundation

- **Stream A:** Draft new introduction, rewrite Section 2 (fix sign error, remove “supply vs demand”, state model closure clearly)
- **Stream B:** Implement η parameter, switch to mobile labor, run pilot sweep
- **Stream C:** Phase R1 (data pipeline), start Phase R2 (core solver)

1.4.2 End of Week 2: Go/No-Go

- Check pilot η sweep results: are sectoral dynamics interesting?
- Check variance decomposition: does η dominate ($\varepsilon, \theta, \sigma$)?
- If yes to either: proceed with full plan
- If no to both: pivot to pure replication strategy (Stream C + Stream A)

1.4.3 Weeks 3–4: Core Results

- **Stream A:** Rewrite Sections 4–5 (CES, labor slack $\rightarrow$ endogenous labor supply), integrate literature
- **Stream B:** Full η sweep + variance decomposition, open economy (or transparent mapping), calibration documentation
- **Stream C:** Phase R2 (solver), Phase R3 (shock simulation)

1.4.4 Weeks 5–6: Integration

- **Stream A:** Write results section (aggregate, sectoral, variance decomposition), discussion, conclusion
- **Stream B:** Generate final figures, update plotting for new results
- **Stream C:** Phase R4 (elasticity gradient), Phase R5 (reallocation) – if time permits

1.4.5 Weeks 7–8: Finalization

- **Stream A:** Copy-edit, fix minor issues, finalize manuscript
- **Stream A:** Write response letter to both reviewers
- **Stream C:** Phase R6 (second-order effects) – if time permits

1.4.6 Week 9–10: Submission

- Final review of all three deliverables
- Submit revised manuscript + response letter + replication code

1.5 Priority Matrix

Task	Stream	Priority	Blocking?	Effort
Implement η parameter	B	Critical	Blocks A (Sec 5–6)	3–5 days
Mobile labor (economy-wide wage)	B	Critical	Blocks η implementation	2–3 days
Variance decomposition	B	Critical	Blocks A (results section)	2–3 days
Pilot η sweep	B	Critical	Go/no-go decision	1 day
New introduction	A	High	–	2 days
Rewrite Section 2 (fixes)	A	High	–	2 days
Literature integration	A	High	–	2–3 days
Open economy	B	Medium	Can be transparent mapping	3–5 days
Calibration documentation	B	Medium	Blocks A (data section)	1 day
Rewrite Sections 4–5	A	High	Needs B results	3–4 days
Results section	A	High	Needs B results	2–3 days
Response letter	A	High	Needs final manuscript	2–3 days
Replication R1–R2	C	Medium	Validates model	5–10 days

Task	Stream	Priority	Blocking?	Effort
Replication R3–R6	C	Low	Nice to have	10–15 days

1.6 Risk Register

Risk	Likelihood	Impact	Mitigation
η sweep results are smooth/monotonic	Medium	High (weakens novelty)	Size claim (variance decomposition) is insurance policy
Reviewer 2 irreconcilable	Medium	High (desk reject)	Maximally gracious response; size claim is genuine contribution
Mobile labor implementation difficult	Low	Medium	Well-understood CGE modification; existing code is modular
Open economy too hard	Medium	Medium	Fallback: transparent documentation of closed-economy mapping
Replication takes too long	Medium	Low	Replication is not blocking; model extension (Stream B) is the critical path

1.7 Reviewer Response Strategy

1.7.1 Reviewer 2 (devastating, recommends rejection)

- Acknowledge core criticism fully: “the bridge already exists” (Robinson 2006)
- Show endogenous labor supply replaces exogenous injection
- Present variance decomposition as genuinely new quantitative finding
- Address every specific point (sign error, closed economy, immobile labor, CES novelty, etc.)
- Be maximally gracious – do not defend the original framing

1.7.2 Reviewer 1 (moderate, constructive)

- Address each point systematically
- Show restructured paper with clear model description
- Include calibration summary table
- Engage with McGregor et al. (1996)
- Clarify aggregate vs sectoral distinction (now quantified via variance decomposition)